

Programming for Artificial Intelligence(2+1)

Course Description:

This course provides students with a strong programming foundation for Artificial Intelligence applications. It focuses on writing efficient, well-structured, and well-documented Python code while introducing students to collaborative development practices using version control systems. The course further equips students with hands-on experience in implementing basic Machine Learning models using standard frameworks and tools. By the end of the course, students will be able to develop reproducible AI solutions, manage AI projects collaboratively, and apply machine learning techniques to real-world datasets.

Course Learning Outcomes (CLOs):

By the end of this course, the student will be able to:

CLO-1: Write efficient and well-documented code.
CLO-2: Utilize version control (Git) to manage and collaborate on AI-related code repositories.
CLO-3: Implement basic Machine Learning models using standard ML frameworks.

CLO-wise Assessment and Weightage

CLO	Assessment Tools	Weightage (%)
CLO-1	Quizzes, Programming Assignments, Mid-Term Examination	40%
CLO-2	Git-based Assignment, Project Repository Evaluation	20%
CLO-3	ML Assignments, Project, Final Examination	40%

CLO–GA Mapping with Justification

CLO	Mapped GA	Justification
CLO-1	GA-4: Design / Development of Solutions	This CLO enables students to design, implement, and document computational solutions following accepted programming standards, supporting the development of effective and maintainable software.
CLO-2	GA-6: Individual and Teamwork	This CLO develops students' ability to work effectively as individuals and team members by using version

		control systems for collaborative software development.
CLO-3	GA-5: Modern Tool Usage	This CLO equips students with hands-on experience in selecting and applying modern machine learning frameworks and tools to solve real-world AI problems.

Weekly Teaching Plan:

Sr. No.	Week-wise & Lecture-wise Topics	CLO(s)
1	Week 01 – L1: Why Python for AI, IDEs, Syntax	CLO-1
2	Week 01 – L2: Variables and Data Types	CLO-1
3	Week 02 – L1: Conditional Statements	CLO-1
4	Week 02 – L2: Loop Statements	CLO-1
5	Week 03 – L1: Lists & Comprehensions	CLO-1
6	Week 03 – L2: Tuples & Dictionaries	CLO-1
7	Week 04 – L1: Functions & Use Cases	CLO-1
8	Week 04 – L2: Lambda, Map, Filter	CLO-1
9	Week 05 – L1: File Handling	CLO-1
10	Week 05 – L2: Exception Handling	CLO-1
11	Week 06 – L1: Classes & Objects	CLO-1
12	Week 06 – L2: OOP Principles	CLO-1
13	Week 07 – L1: ML Overview & Types	CLO-3
14	Week 07 – L2: Sklearn & Pandas	CLO-3
15	Week 08 – L1: Data Cleaning	CLO-3
16	Week 08 – L2: Matplotlib Visualization	CLO-3
17	Mid-Term Examination	—
18	Week 09 – L1: Seaborn I	CLO-3
19	Week 09 – L2: Seaborn II	CLO-3
20	Week 10 – L1: KNN & Logistic Regression	CLO-3
21	Week 10 – L2: Evaluation Metrics	CLO-3
22	Week 11 – L1: Linear & Polynomial Regression	CLO-3
23	Week 11 – L2: Regression Metrics	CLO-3
24	Week 12 – L1: Decision Trees	CLO-3
25	Week 12 – L2: Random Forest & Explainability	CLO-3
26	Week 13 – L1: Git Basics	CLO-2
27	Week 13 – L2: GitHub Workflow	CLO-2
28	Week 14 – L1: Deploying ML App with streamlit	CLO-2
29	Week 14 – L2: Web Scrapping in Python	CLO-1
30	Week 15 – L1: Web Scrapping in Python	CLO-1
31	Week 16 – L1: Databases in Python	CLO-1
32	Week 16: Project Presentations and Revision	CLO-1, CLO-2, CLO-3
33	Final Examination	

Recommended Textbooks:

1. Introduction to Machine Learning with Python” — Andreas C. Müller & Sarah Guido
2. **“Fluent Python” — Luciano Ramalho**
3. *Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow* by Aurélien Géron.